



dr MILENA ŠOŠIĆ
mathematics & programming



GENERAL:

Machine Learning expert with over twenty years of experience in software engineering and machine learning algorithm design and implementation. Proven track record of delivering results in large-scale projects across finance, telecom, and commercial industries. PhD degree in Natural Language Processing (NLP) field.

OBJECTIVE:

To increase my professional career in the field of machine learning and software development.

EDUCATION:

**Faculty of Mathematics
University of Belgrade
Computer Science
2020 – 2025**

PhD degree in computer science and NLP. Dissertation titled: *“Modeling moral and emotional language aspects in conversational texts classification”*. Research interest include semantic text analysis, Information Retrieval and Text Classification through the integration of Machine Learning and Knowledge-Based approaches, particularly the use of Knowledge Graphs for entity extraction, linking, and semantic representation learning.

**Faculty of Mathematics
University of Belgrade
Computer Science
2004 – 2010**

Magister degree in Computer Science, thesis: *“Applying classification methods on N-gram genome analysis”*, average grade **10** out of 10, specialization in bioinformatics and machine learning.

**Faculty of Mathematics
University of Belgrade
Computer Science**

University of Belgrade, Graduated Mathematician for Computer Science, average grade **8,86** out of 10.0.

1997 – 2004

- **Study Programs in Computer Science:** Programming Languages (Pascal, C, C++, Java, Prolog, Lisp), Algorithms Architecture, Computer System Architecture, Mathematical Logic in Computer Science, Automata Theory, Databases, Microprocessors, Microcomputers, Assembly Language, Compilers and Interpreters, Programmable Systems Architecture.
- **Study Programs in Mathematics:** Algebra (I and II), Analytical, Constructive and Euclidian Geometry, Calculus (I and II), Real and Complex Functions, Numerical Methods (I and II), Probability and Statistics.

High school “Požarevačka Gimnazija”
Department for mathematics
Music school “Stevan Mokranjac”
Department for flute
1993 – 1997

Excellent student, “Vuk Karadžić” certificate.

Musical performer – flutist.

PUBLICATIONS & CONFERENCES:

- [*Building an Emotion Lexicon for Serbian Using Curated Language Resources*](#), Language Resources and Evaluation, Springer, 2025.
- [*Effective Methods for Email Classification: Is it a Business or Personal Email?*](#), Computer Science and Information Systems, 2022.
- [*SRPOL - A Lexicon Based Framework for Sentiment Strength of Serbian Texts*](#), Review of the National Center for Digitization, 2022.
- presenter, [*Artificial Intelligence Conference*](#), SANU & Mathematical Institute, Belgrade, 2023/2025.
- presenter, [*South Slavic Languages in the Digital Environment Conference*](#), JeRTeh, Belgrade, 2024.

MEMBERSHIP:

JeRTeh – multidisciplinary academic society dedicated to the development of language resources and technologies for Serbian language.

COMPUTER SKILLS:

programming languages:

Python, C#, C++, C, Java, JavaScript, SQL.

databases:

MS SQL Server, Oracle, IBM DB2, PostgreSQL, MongoDB, Neo4j, Redis, FAISS, Milvus, Azure Search Index, Vertex AI Search.

machine learning:

TensorFlow, Keras, PyTorch, Scikit-learn, DSPy, AutoML, MLflow, Jupyter, Pandas, NumPy, Matplotlib, Seaborn.

NLP:

HuggingFace & Sentence Transformers, spaCy, NLTK, Genism.

LLMs, RAG & Evaluation:	LLM APIs, Ollama, LlamaIndex, LangFuse, Ragas, DeepEval, wandb.
Orchestration & Agents:	LangGraph, LangChain, Promptflow.
Cloud platforms:	Azure, GCP.
courses:	• Machine Learning, Stanford, Coursera. • Deep Learning Specialization (5 courses), Coursera. • Natural Language Processing Specialization (4 courses), Coursera. • Learn Python Programming Masterclass, Udemy. • Understanding and Visualizing Data with Python, Coursera. • ITIL V3 Foundations (IT Service Management). • Introduction to R, edX. • Honor Code Certificate for Programming with C#, edX.

WORKING EXPERIENCE:

March 2024 – current Senior Data Scientist Publicis Sapient	Work responsibilities include: • Leading a team of data scientists in developing advanced NLP solutions for real-world business applications. • Designing and implementing systems based on LLMs, RAG pipelines, and agent-based architectures. • Applying graph-based approaches and knowledge graphs for semantic representation, entity linking, and structure-aware information retrieval. • Collaborating cross-functionally with engineering and product teams to deliver scalable, production-grade AI solutions. • Driving research and experimentation to evaluate and integrate state-of-the-art language models and AI frameworks into client projects.
January 2021 – March 2024 Senior Data Scientist GridDynamics	Work responsibilities to: • Mentor data science interns and interview candidates. • Develop PoC solutions for R&D projects by exploring the most advanced AI techniques and tools. • Design and implement new solutions with focus on predictive analytics. • Run projects in NLP domain using newest machine learning technologies.
February 2019 – January 2021 AI Team Lead Devtech	Work activities related to: • Lead AI team, mentor and drive AI efforts. • Develop PoC solutions for existing and past projects by adding AI components. • Design and implement new solutions with focus on predictive analytics. • Run projects in text and sequence analysis domain using newest machine learning technologies.

February 2020 – March 2022 Advisory Board Member Serbian AI Society	Serbian AI Society is being launched in Serbia by the most prominent people in artificial intelligence, with one goal - networking and strengthening the domestic AI scene. The aim of this organization is to bring together all stakeholders dealing with this subject which will contribute to the presence and growth of artificial intelligence in Serbia
January 2016 – February 2019 Machine Learning Team Lead Selerant	Work on the projects in the PLM area, in the machine learning field, especially: • Lead a team of software developers. • Responsible for business intelligence applications and systems. • Manage projects using agile methodology. • Manage state-of-the-art machine learning tasks for text segmentation and classification. • Shape product features according to the market requirements. • Cross-functional communication and coordination.
December 2013 – January 2016 Data Scientist Telenor	Work on the high-visibility projects in the banking and telecommunication areas. Utilize analytical tools like IBM SPSS and STATISTICA to identify trends and relationships between different pieces of data, draw appropriate conclusions and translate analytical findings into risk management and marketing strategies that drive value. Especially: • Develops different up-sell, cross-sell and churn models. • Supports risk models development. • Accountable for segmentation, customer lifetime value calculation. • Supports campaign launches by data preparation, extraction and reporting of results. • Supports customer data mart development.
December 2009 – December 2013 Solutions Analyst PepsiCo	Manage software solutions within the company with focus on SAP change improvement process, precisely taking following responsibilities: • SAP change management and analysis in the FICO/COMM/OPS/BI domains. • Functional Specification Design. • Business process analysis and optimization. • SAP security and control activities.
January 2007 – December 2009 Senior software developer in .NET Dominion Voting Systems Corporation	Maintain, develop and manage different types of software solutions for electronic voting purpose using .NET tools and techniques

August 2004 – January 2007
Software developer in .NET
SDD ITG

Develop new software solutions using .NET technology and C# programming language tools with main focus on software development for RFID readers.

LANGUAGES:

English – fluent, written and spoken.
Serbian – mother-tongue.

PERSONAL CHARACTERISTICS:

Highly organized and reliable professional with strong analytical and problem-solving skills. Recognized for clear communication, thorough documentation, and efficiency in meeting project requirements. Hard-working, responsible, and motivated to deliver high-quality results.

HOBBIES:

Hiking, reading, playing flute.